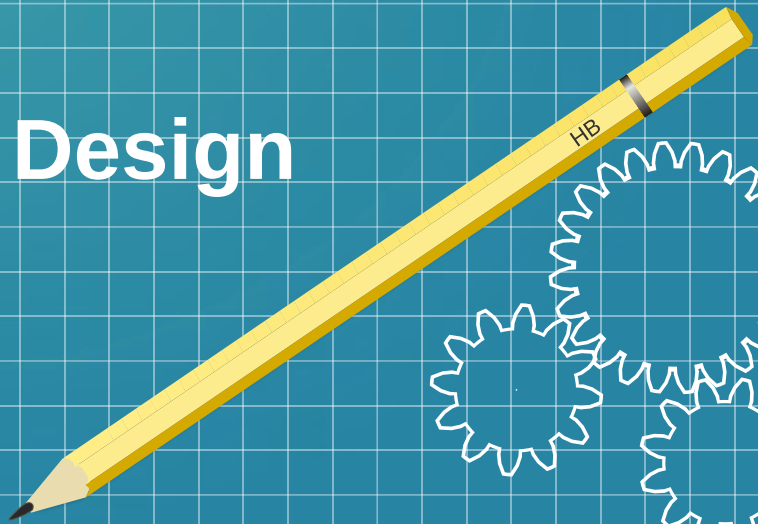
CSC-300-Software-Engineering

Unit–III: **User Interface Design**



User Interface

- The user interface part of a software product is responsible for **all interactions with the end-user**
- An **interface that is difficult to use leads to higher levels of user errors** and ultimately leads to **user dissatisfaction**
- For many interactive applications, as much as **50 per cent of the total development effort** is spent on developing the user interface part

Characteristics of a good user interface

- **Speed of learning**
  - A good user interface should be **easy to learn**
  - Speed of learning is **hampered by complex syntax** and **semantics of the command issue procedures**
  - A good user interface should **not require its users to memorise commands**
  - The user be **should not be asked to remember information from one screen to another**

- **Use of metaphors** and **intuitive command names**
  - For example - user interface of a text editor uses concepts similar to the tools used by a writer
- **Consistency**
  - User should be able to use the similar commands in different circumstances for carrying out similar actions
- **Component-based interface**
  - Developed using some standard user interface components

- **Speed of use**
  - **The time and user effort necessary to initiate and execute different commands should be minimal**
- **Speed of recall**
  - Once users learn how to use an interface, **the speed with which they can recall the command issue procedure should be maximised**

- **<u>Error Prevention</u>**:
  - A good user interface should **minimise the scope of committing errors while initiating different commands**
- **<u>Aesthetic and attractive</u>**:
  - A good user interface should be **attractive to use**

- **Feedback**:
  - A good user interface must provide **feedback to various user actions**
- **Support for multiple skill levels:**
  - A good user interface should **support multiple levels of sophistication of command issue procedure** for different categories of users

- **Error recovery (undo facility)**:
  - A good user interface should **allow a user to undo a mistake committed by him** while using the interface
- **User guidance and on-line help**:
  - Whenever users need guidance or seek help from the system, they should be **provided with appropriate guidance and help**

Mode-based vs Modeless Interface

- **A mode is a state or collection of states in which only a subset of all user interaction tasks can be performed**
- In a **mode-based interface**, **different sets of commands can be invoked depending on the mode in which the system is**
- In a **modeless interface, the same set of commands can be invoked at any time during the running of the software**
- A **modeless interface has only a single mode**

- A mode-based interface **can be represented using a state transition diagram**, where **each node of the state transition diagram would represent a mode**

Graphical User Interface (GUI) vs Text-based User Interface

- **GUI allows multiple windows with different information to be displayed simultaneously on the user screen.** This gives **the flexibility to simultaneously interact with several related items at any time** and **can have access to different system information**.
- **Iconic information representation and symbolic information manipulation is possible in a GUI**

- **GUI usually supports command selection using an attractive and user-friendly menu selection system**
- **GUI enables using a pointing device such as a mouse or a light pen to be used for issuing commands**
- **GUI requires special terminals with graphics capabilities for running and also requires special input devices such a mouse**. On the other hand, **a text-based user interface can be implemented even on a cheap alphanumeric display terminal**